SYS 304: Trade Studies, Risk, and Decision Analysis

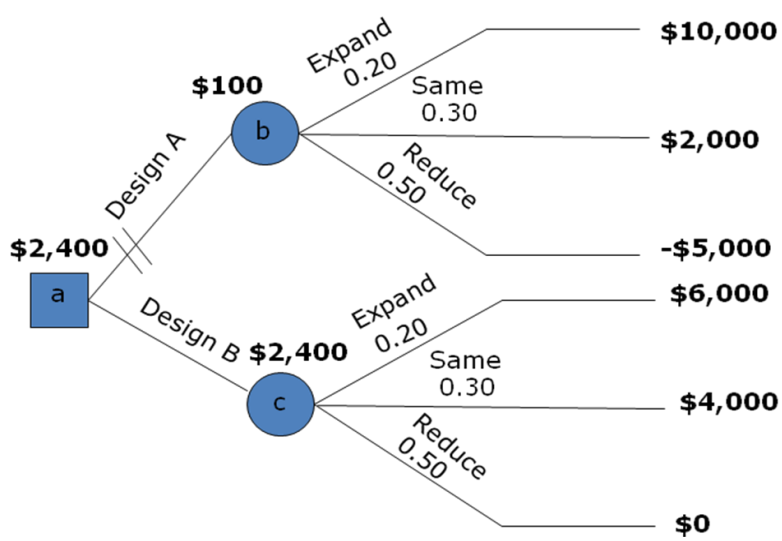
Decision Making Under Risk and Uncertainty

Homework 3 Solutions

1. A systems engineering firm has two design options to choose from to meet a contractual agreement with another company (customer): Design A and Design B. The return that will be achieved by each of these depends on whether the customer operations are likely to expand in the future, stay the same, or reduce in the future. The payoff table for this scenario is given below.

Alternative	State of Nature / Probability		
	Expand 0.2	Same 0.3	Reduce 0.5
Design A	$10,000	$2,000	-$5,000
Design B	$6,000	$4,000	$0

Draw the decision tree to represent this situation and perform the necessary calculations to determine which one of the two designs is better. Which one should you choose to maximize thee expected value?



- *b*: EP = (0.2)($10,000) + (0.3)($2,000) + (0.5)(–$5,000) = $100
- *c*: EP = (0.2)($6,000) + (0.3)($2,000) +(0.5)($0) = $2,400

Design B should be selected.

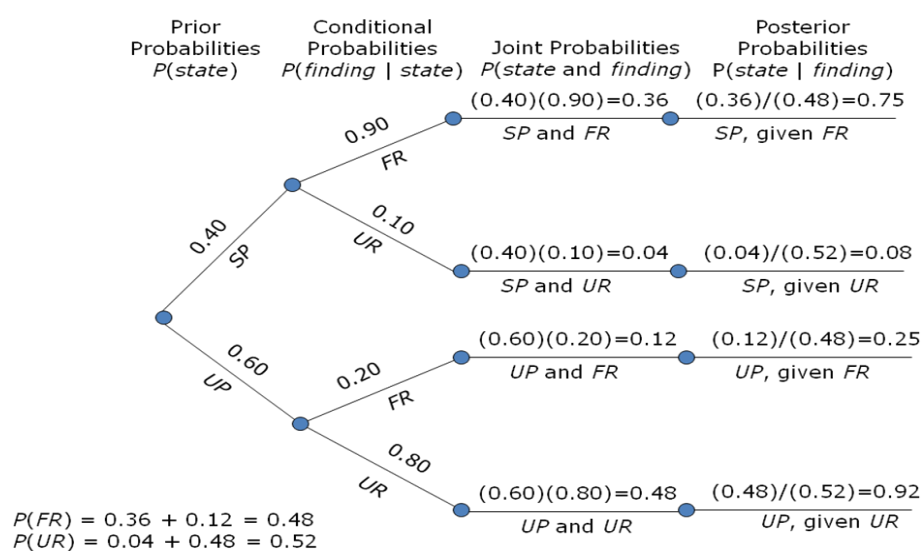
2. A start-up company has plans to develop a new software application. However, the company believes that the chances of success for the application are only about 40%. A contractor has suggested that the company conduct a survey in the community to get a better feeling of the demand for the application. There is a 0.9 probability that the research will be favorable if the product is successful. Furthermore, it is estimated that there is a 0.8 probability that the marketing research will be unfavorable if the product is unsuccessful. Determine the chances of a successful product given a favorable result from the marketing survey.

Use Bayes' theorem:
$P(SP)$ = probability of successful product
$P(UP)$ = probability of unsuccessful product
$P(FR \mid SP)$ = probability that the research will be favorable given a successful product
$P(UR \mid SP)$ = probability that the research will be unfavorable given a successful product
$P(UR \mid UP)$ = probability that the research will be unfavorable given an unsuccessful product
$P(FR \mid UP)$ = probability that the research will be favorable given an unsuccessful product

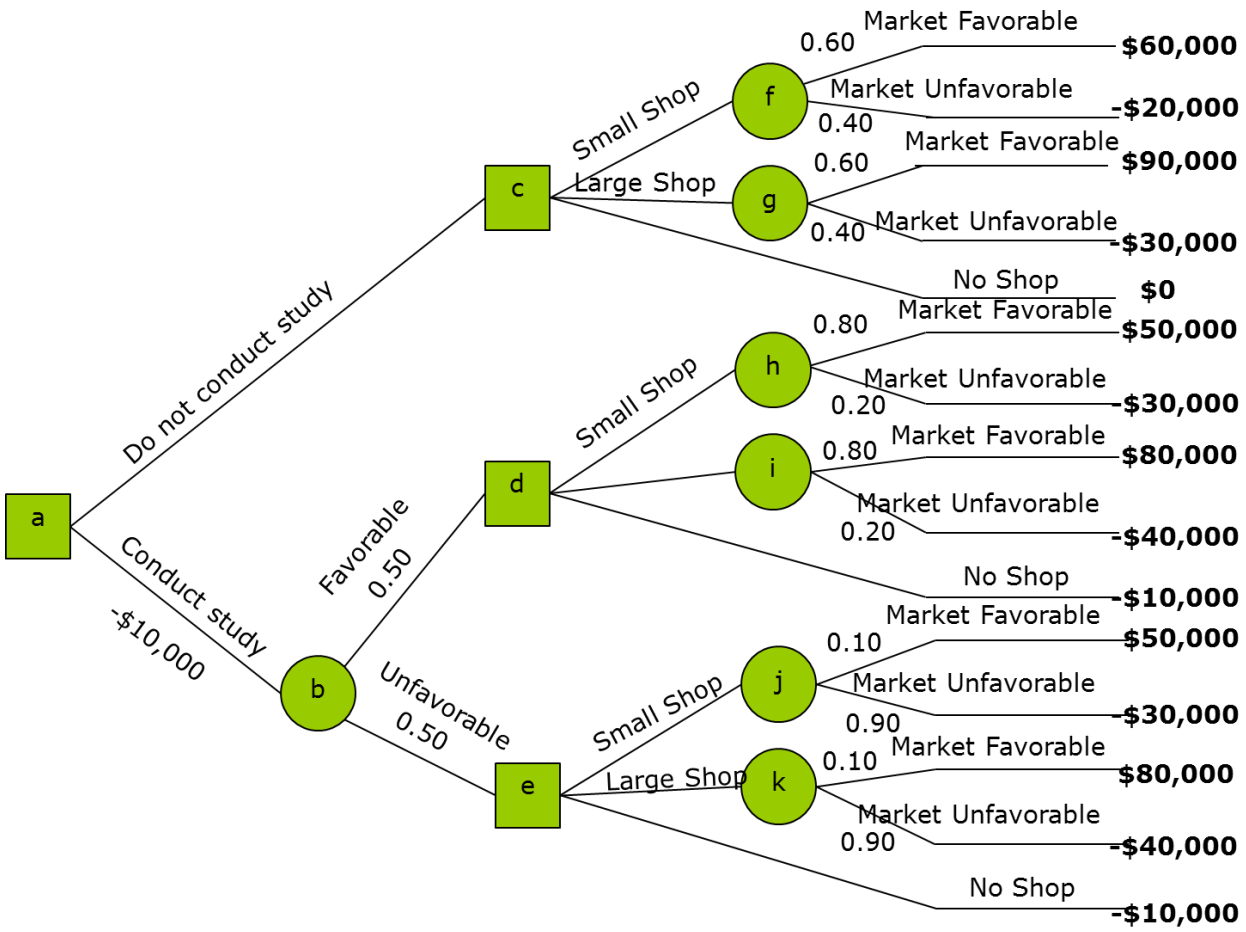
Given:
$P(SP) = 0.4$
$P(FR \mid SP) = 0.9$
$P(UR \mid UP) = 0.8$

Calculate:
$P(UP) = 1 - P(SP) = 1 - 0.4 = 0.6$
$P(UR \mid SP) = 1 - P(FR \mid SP) = 1 - 0.9 = 0.1$
$P(FR \mid UP) = 1 - P(UR \mid UP) = 1 - 0.8 = 0.2$

Bayes' Theorem:
$P(SP \mid FR) = [P(FR \mid SP)][P(SP)] / \{[P(FR \mid SP)][P(SP)] + [P(FR \mid UP)][P(UP)]\}$
$P(SP \mid FR) = [(0.9)(0.4)] / [(0.9)(0.4) + (0.2)(0.6)]$
$P(SP \mid FR) = (0.36) / (0.36 + 0.12) = (0.36) / (0.48) = 0.75$



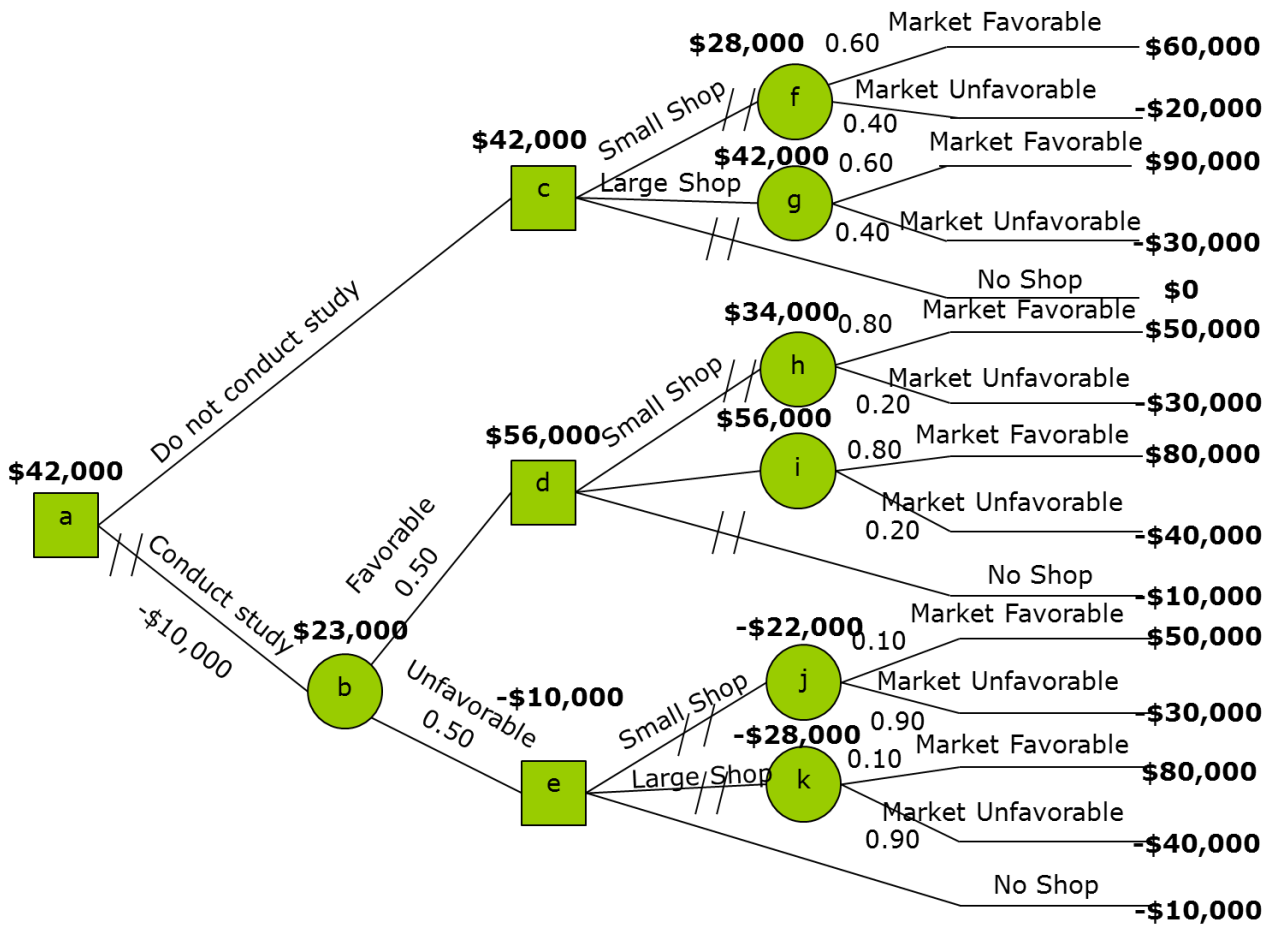
3. A hardware development company considers launching a new product computer hardware. Because of the initial investment needed the company has decided to conduct a pilot study to assess the market for the product. The pilot study cost is $10,000, and it can be either successful or not successful. The company's main decision is whether to build a large manufacturing facility, a small manufacturing facility, or no facility at all. With a favorable market, the company expects to make $90,000 from the large facility or $60,000 from the smaller facility. If the market is unfavorable, however, the company estimates a loss of $30,000 with the large facility, and a loss of $20,000 with the small facility. The probability of a favorable market given a successful pilot is 0.8. The probability of an unfavorable market given an unsuccessful pilot study result is estimated to be 0.9. The company believes that there is a 50-50 chance the pilot study will be successful. The company could bypass the pilot study and simply make the decision as to whether build a large plant, small plant, or no facility at all. Without doing any testing in the pilot study, the company estimates that the probability of a successful market is 0.6. Analyze the decision problem and provide your recommendation.



- *f*: EP = (0.6)($60,000) + (0.4)(-$20,000) = $28,000
- *g*: EP = (0.6)($90,000) + (0.4)(-$30,000) = $42,000
- *h*: EP = (0.8)($50,000) + (0.2)(-$30,000) = $34,000

- *i*: EP = (0.8)($80,000) + (0.2)(-$40,000) = $56,000
- *j*: EP = (0.1)($50,000) + (0.9)(-$30,000) = -$22,000
- *k*: EP = (0.1)($80,000) + (0.9)(-$40,000) = -$28,000

- *c*: EP(No Shop) = $0 < EP(Small Shop) = $28,000 < EP(Large Shop) = $42,000
  - Choose Large Shop.
- *d*: EP(No Shop) = -$10,000 < EP(Small Shop) = $34,000 < EP(Large Shop) = $56,000
  - Choose Large Shop.
- *e*: EP(Large Shop) = -$28,000 < EP(Small Shop) = -$22,000 < EP(No Shop) = -$10,000
  - Choose No Shop



- *b*: EP = (0.5)($56,000) + (0.5)(-$10,000) = $23,000

- *a*: EP(Conduct study) = $23,000 < EP(Do not conduct study) = $42,000
  - Choose Do not conduct study.

Decision: Do not conduct study and Large Shop with EP = $42,000